Reconstruction and Recognition of Tensor-Based Objects With Concurrent Subspaces Analysis

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***Abstract*—Principal Components Analysis (PCA) has traditionally been utilized with data expressed in the form of 1-D vectors, but there exists much data such as gray-level images, video sequences, Gabor-filtered images and so on, that are intrinsically in the form of second or higher order tensors. For representations of image objects in their intrinsic form and order rather than concatenating all the object data into a single vector, we propose in this paper a new optimal object reconstruction criterion with which the information of a high-dimensional tensor is represented as a much lower dimensional tensor computed from projections to multiple concurrent subspaces. In each of these subspaces, correlations with respect to one of the tensor dimensions are reduced, enabling better object reconstruction performance. Concurrent subspaces analysis (CSA) is presented to efficiently learn these subspaces in an iterative manner. In contrast to techniques such as PCA which vectorize tensor data, CSA's direct use of data in tensor form brings an enhanced ability to learn a representative subspace and an increased number of available projection directions. These properties enable CSA to outperform traditional algorithms in the common case of small sample sizes, where CSA can be effective even with only a single sample per class. Extensive experiments on images of faces and digital numbers encoded as second or third order tensors demonstrate that the proposed CSA outperforms PCA-based algorithms in object reconstruction and object recognition.**

***Index Terms*—Concurrent subspaces analysis (CSA), dimensionality reduction, object reconstruction, object representation, principal components analysis (PCA).**

I. Introduction

THE MOST popular algorithm for dimensionality reduction is Principal Components Analysis (PCA) [4], [18], which searches for a most representative subspace for data representation [24]. Among the applications in which PCA has been successfully used are data compression and face recognition.

PCA has traditionally been utilized with data expressed in the form of 1-D vectors, but there exists much data such as image objects that are intrinsically in the form of second or higher order tensors [1]. For example, gray-level images represent second order tensor data (a matrix), and can be expanded to a third order tensor with the inclusion of temporal data, such as video sequences in content analysis [3], or by representing sets of Gabor filter images [5], [25] as often used in face recognition. In these third-order tensors, the first two dimensions represent the image height and the image width, and the third dimension represents the temporal dimension or the Gabor feature, respectively. To process higher order data with PCA, the data is converted to 1-D vectors before analysis, which we refer to as an *image-as-vector* representation in this work. The dimensionality of the concatenated 1-D data in PCA is usually very high, while in real object recognition applications such as face recognition the available number of training samples is small. In consequence, subspaces learnt from PCA may inadequately represent a sample distribution, resulting in poor recognition performance [15].

To avoid this problem, it is often helpful to process the data in its original form and order. In the literature, several unsupervised techniques [15], [26], [27], [30], [31] have been proposed for good reconstruction of data with matrix representations, which we refer to as an *image-as-matrix* representation in this work. Shashua *et al.* [15] applied rank-1 decomposition for object reconstruction of second order tensor data. In [30], Yang *et al.* proposed a scheme called *2DPCA* to conduct dimensionality reduction with a 2-D matrix representation. In [26], we proved that 2DPCA can be considered as a special PCA in which row vectors are the analyzed objects. In this way, 2DPCA considers correlations only among row vectors and ignores correlations among column vectors; moreover, it requires a larger number of coefficients than PCA for image representation, as the authors of [30] have themselves indicated. In [26], we proposed a two-step scheme called *Parallel Image Matrix Compression* (PIMC) to conduct PCA with the object feature first as row vectors of the image and then as column vectors, thus removing redundancy among both rows and columns. We further proposed a new scheme, namely *Coupled Subspace Analysis* (CSA-2) [27], in which rank-(R1, R2) decomposition is applied with an objective function similar to [15]. As shown in [27], from an algorithmic viewpoint, PIMC is a special case of CSA-2 with the iteration number set to one. We note that Ye *et al.* [31] also proposed an approach called *Low Rank Approximation of Matrices* (LRAM) that is identical to CSA-2, i.e., applies rank-(R1, R2) decomposition for a good object reconstruction with a 2-D matrix representation. The 2DPCA algorithm was shown be a special case of [31] and [27].

[1]Note that here the tensor is used in a context different to the mathematical definition of a tensor.

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In this work, we generalize our previous approaches [26], [27] to process image objects in the form of general high-order tensors, which have proven to be effective in face recognition [25], digital number recognition [5], and content analysis [3]. This work was initially published in [28]. For image objects represented as second– or higher order tensors, we propose an optimal reconstruction criterion for general high-order tensor objects, where the information of a higher dimensional tensor is encoded as a much lower dimensional tensor by projections to multiple concurrent subspaces. In each subspace, redundancies and correlations among one of the tensor dimensions are removed. With the recently developed mathematics of higher order tensor decomposition [7], [9], [10] we propose an iterative approach, called CSA or CSA-3, to seek an optimal solution for the objective function. In each step of CSA, optimization of the reconstruction criterion is formulated as a specialized PCA, where the object tensor is first unfolded into a matrix along the dimension to be analyzed, and then PCA is conducted with respect to the column vectors of the unfolded matrices.

The use of multiple concurrent subspaces with tensor-based objects leads to clear advantages in dimensionality reduction. Since the correlations within each tensor dimension are reduced, the size of the feature space can be more effectively decreased than in methods where these correlations are not fully considered. With a reduced number of features, reconstruction performance is enhanced, and the ability to learn a representative subspace is improved, especially in cases with little training data. A more representative subspace directly leads to better object recognition. Performance is further elevated because of the relatively large number of available projection directions in CSA in comparison to most traditional algorithms. Additionally, the reduction in data dimensionality brings appreciable reductions in computational costs because of the fewer data dimensions that need to be learned for each subspace.

Higher order tensor decomposition [7], [9], [10] has become popular in computer vision [2], [6], [17], [23], as pioneered by the Tensorface method [19], [20], [21]. Tensorface represents a 2-D gray-level image as a 1-D vector and organizes the image ensembles of different persons under different pose, illumination and expression into a high-order tensor. From an algorithmic viewpoint, Tensorface and CSA are both variants of N-mode SVD (a.k.a. rank-$(\mathrm{R1}, \mathrm{R2}, \ldots, \mathrm{RN})$ decomposition or high-order SVD) [8], [10], [22]. However, CSA and Tensorface are intrinsically different because of the following reasons. 1) Tensorface specifically deals with gray-level image input, which is converted to a 1-D vector, and thus it can be classified as an *image-as-vector* representation. Tensorface aims to characterize the variations caused by *external* factors, such as pose, illumination and expression. We also note that Tensorface does not address Gabor-filtered images or video sequences as input; it is unclear how Tensorface can deal with such image data. In CSA, we directly represent one gray-level image as a second order tensor (a matrix), and one Gabor filtered image or one video sequence as a third order tensor, thus it can be considered as an *image-as-matrix* representation. CSA aims to use several subspaces to characterize the *internal* variations of an object, such as of rows, columns, and Gabor features. 2) Tensorface requires a large number of samples of different pose, illumination and expression for training, which is generally unavailable in real applications such as face recognition. *When the number of images per person is only one or the total number of images is a prime, Tensorface is strictly equivalent to PCA for gray-level image input*. In contrast, when each person has only a limited number of images or even just a single image, CSA is more effective than PCA, as demonstrated in the experiments. 3) Potentially, CSA could be extended to address both internal and external factors by constructing tensors with the first several dimensions representing width, height, Gabor features, etc., and the other dimensions encoding additional factors such as pose, illumination, and expression.

While *image-as-matrix* representations were used in [2], [6], [17], these works aimed at supervised classification or manifold learning, without targeting optimal object reconstruction. Another recent work [12] systematically compared several different tensor decomposition techniques and reported that Parallel Factors Decomposition or Canonical Decomposition achieved better performance than the classical N-mode SVD in content-based indexing and retrieval. The methods of [2], [6], [12], [17] were published after our conference paper on this work [28]. Wang and Ahuja [23] concurrently proposed a scheme with images in a matrix representation to approximate high-order tensor input; in face recognition applications, they consider only 2-D gray-level input. We note that 2-D or 3-D wavelet transforms [29] are frequently used for compressing images or video sequences. The main difference between wavelet compression and CSA compression is that the former reconstructs a single image with a fixed basis, while CSA compression uses a data-dependent basis that aims to minimize correlations among different features in a set of images.

The remainder of this paper is organized as follows. In Section II, we introduce the optimal reconstruction criterion for general high-order tensor objects, and then present an iterative algorithm and algorithmic analysis of CSA for the optimization of the new criterion. The relationship of CSA to existing algorithms is discussed in Section III. Then experiments on object reconstruction and object recognition are reported, followed by conclusions.

II. Concurrent Subspaces Analysis

After briefly reviewing background material, we propose the optimal reconstruction criterion for general high-order tensor objects, and then present an iterative algorithm to acquire an optimal solution. We also investigate issues including learnability, computational complexity, and objective function convergence.

A. Background

The basic objective of dimensionality reduction is to learn a projection matrix that transforms the original high-dimensional data to a lower dimension. In Fig. 1, the three rows illustrate the original high-dimensional object, the lower dimensional object, and the reconstructed object, respectively. Let us represent the training images by second order tensors (matrices): $X_i \in \mathbb{R}^{m \times n}, i = 1, \ldots, N$, where m and n denote the height and the width of the original images, respectively, and N be the total number of images. We also denote $x_i \in \mathbb{R}^{mn}$ as the corresponding vector representation for the ith image X_i.

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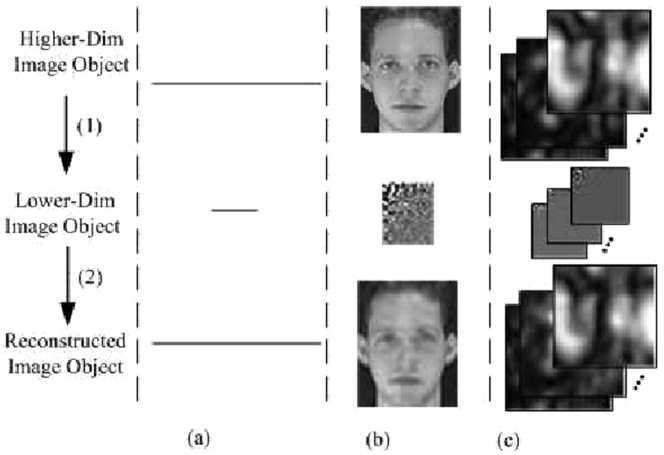


Fig. 1. Object dimensionality reduction and reconstruction for vectors, matrices, and higher order tensors. Note that (1) indicates dimensionality reduction, and (2) signifies object reconstruction.

In Fig. 1(a), Principal Components Analysis (PCA) [18] is applied for subspace learning with each image object encoded as a 1-D vector that is formed by concatenating image rows. The objective function of PCA is expressed as

$$P^* = \arg\min_P \sum_{i=1}^{N} \|PP^T x_i - x_i\|^2 \tag{1}$$

where $P \in \mathbb{R}^{mn \times d}$ is the pursued projection matrix, and d is the dimension after reduction. Eigenvalue decomposition can be used directly for acquiring the solution.

In [15] and [31] as well as our previous work [27], an image object is directly represented as a matrix as illustrated in Fig. 1(b). The objective function in [15] is as follows:

$$(u_j|_{j=1}^d, v_j|_{j=1}^d)^* = \arg \min_{u_j|_{j=1}^d, v_j|_{j=1}^d} \sum_{i=1}^{N} \| \sum_{j=1}^{d} u_j u_j^T X_i v_j v_j^T - X_i\|^2 \tag{2}$$

where d is the lower dimensionality, $u_j|_{j=1}^d$ stands for $u_1, \ldots, u_d, v_j|_{j=1}^d$ denotes $v_1, \ldots, v_d, u_j \in \mathbb{R}^m$, and $v_j \in \mathbb{R}^n$ are two projection *vectors* with $\|u_j\| = 1$ and $\|v_j\| = 1$. So for each input matrix X, the low-dimensional representation is $Y = [y_1, \ldots, y_d]$, with $y_j = u_j^T X v_j$.

In contrast, the objective function in [27] and [31] is

$$(U, V)^* = \arg\min_{U,V} \sum_{i=1}^{N} \|UU^T X_i VV^T - X_i\|^2 \tag{3}$$

where $U \in \mathbb{R}^{m \times m'}$ and $V \in \mathbb{R}^{n \times n'}$ are two projection *matrices*, and m' and n' denote the height and the width of the low-dimensional images, respectively. So for each input matrix X, the low-dimensional representation is $Y = U^T X V$.

While an image-as-matrix representation was used in [15], [27], and [31], they represent two different ways to approximate original tensors with tensors of lower rank. More specifically speaking, the work in [15], applied a rank-1 decomposition, while the work in [27] and [31] used a rank-(R1, R2) decomposition. Note that in (2), u_d and v_d are projection *vectors* (i.e., the ranks of the two vectors are 1), and U and V in (3) are two projection matrices. Since no closed-form solution exists for the objective functions (2) and (3), the methods in [15], [27], and [31] propose to use iterative schemes to acquire an optimal solution. For further details, we refer the reader to [15], [27], and [31].

In this work, we investigate how to conduct dimensionality reduction by preserving the most representative information in cases where objects are encoded as tensors of arbitrary order. In Fig. 1(c), Gabor filtered images represented as third order tensors are used for illustration. Before formally introducing our work, some terminology on tensor operations are given as follows, with further details available in [7], [9], and [10].

Definition 1: (*tensor inner product, norm and distance*). The inner product of two tensors $\mathbf{X} \in \mathbb{R}^{m_1 \times m_2 \times \ldots \times m_n}$ and $\mathbf{Y} \in \mathbb{R}^{m_1 \times m_2 \times \ldots \times m_n}$ is defined as $\langle \mathbf{X}, \mathbf{Y} \rangle = \sum_{i_1=1,\ldots,i_n=1}^{m_1,\ldots,m_n} \mathbf{X}_{i_1,\ldots,i_n} \mathbf{Y}_{i_1,\ldots,i_n}$. The norm of tensor $\mathbf{X}$ is therefore defined to be $\|\mathbf{X}\| = \sqrt{\langle \mathbf{X}, \mathbf{X} \rangle}$, and the tensor distance between tensors $\mathbf{X}$ and $\mathbf{Y}$ is computed as $D(\mathbf{X}, \mathbf{Y}) = \|\mathbf{X} - \mathbf{Y}\|$.

Definition 2: (*mode-k product*). The mode-k product of tensor $\mathbf{X}$ with matrix $U \in \mathbb{R}^{m'_k \times m_k}$, i.e., $\mathbf{Y} = \mathbf{X} \times_k U$, is defined as $\mathbf{Y}_{i_1,\ldots,i_{k-1},i,i_{k+1},\ldots,i_n} = \sum_{j=1}^{m_k} \mathbf{X}_{i_1,\ldots,i_{k-1},j,i_{k+1},\ldots,i_n} U_{i,j}, i = 1, \ldots, m'_k$.

Definition 3: (*mode-k flattening*). The mode-k flattening of an nth order tensor $\mathbf{X} \in \mathbb{R}^{m_1 \times m_2 \times \ldots \times m_n}$ into a matrix $X^k \in \mathbb{R}^{m_k \times \prod_{i \neq k} m_i}$, i.e., $X^k \Longleftarrow_k \mathbf{X}$, is defined as $X^k_{i_k,j} = \mathbf{X}_{i_1,\ldots,i_n}, j = 1 + \sum_{l=1,l\neq k}^{n} (i_l - 1) \prod_{o=l+1,o\neq k}^{n} m_o$.

B. Optimal Reconstruction Criterion for General Higher Order Tensor Objects

While previous works [15], [26], [27], [30], [31] have dealt with 2-D matrix input of gray-level features, the Gabor filtered images or video sequences frequently used in object recognition and data compression are in the form of third order tensors. In face recognition and digital number recognition, Gabor features [5], [25] have brought much higher levels of performance, where the third dimension indexes Gabor features, i.e., the responses from different Gabor filters. In this work, we investigate how to preserve the most representative information of high-order tensor data. In the following discussion, we denote the training images as nth order tensors $\mathbf{X}_i \in \mathbb{R}^{m_1 \times m_2 \times \ldots \times m_n}, i = 1, \ldots, N$ with N as the total number of images. We also express the low-dimensional representation for a sample $\mathbf{X}$ as $\mathbf{Y}_i \in \mathbb{R}^{m'_1 \times m'_2 \times \ldots \times m'_n}, m'_k < m_k, k = 1, \ldots, n$, the reconstructed tensor as $\mathbf{X}_i^R \in \mathbb{R}^{m_1 \times m_2 \times \ldots \times m_n}$, and the n projection matrices as $U_1, \ldots, U_n$ with $U_k \in \mathbb{R}^{m_k \times m'_k}$.

With projection matrices $U_k|_{k=1}^n$, the low-dimensional representation $\mathbf{Y}_i$ can be acquired as $\mathbf{Y}_i = \mathbf{X}_i \times_1 U_1^T \ldots \times_n U_n^T, i = 1, \ldots, N$. The reconstructed tensor $\mathbf{X}_i^R$ can be obtained as $\mathbf{X}_i^R = \mathbf{Y}_i \times_1 U_1 \ldots \times_n U_n$, so

$$\mathbf{X}_i^R = \mathbf{X}_i \times_1 U_1 U_1^T \ldots \times_n U_n U_n^T. \tag{4}$$

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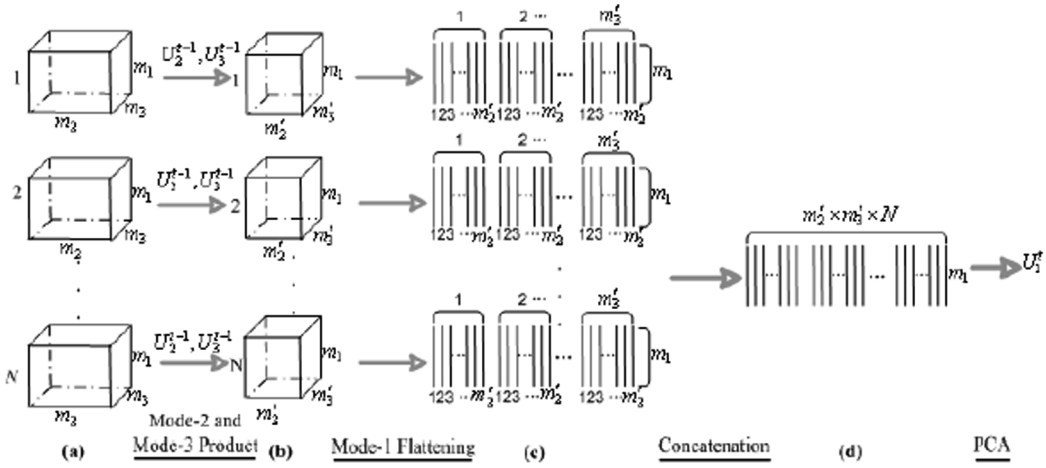


Fig. 2. Flowchart of one step of CSA in the case of a third order tensor.

The projection matrices $U_k|_{k=1}^n$ can be computed based on the *Optimal Reconstruction Criterion for General Higher Order Tensor Objects*

$$(U_k|_{k=1}^n)^* = \arg\min_{U_k|_{k=1}^n} \sum_{i=1}^{N} \|\mathbf{X}_i \times_1 U_1 U_1^T \ldots \times_n U_n U_n^T - \mathbf{X}_i\|^2. \quad (5)$$

The relationship between (5) and (3) will be analyzed in Section III-C. The sum in (5) describes the reconstruction error from N samples (or images). To the best of our knowledge, there exists no closed-form solution for this objective function. Inspired by recent advances in higher order tensor decomposition [7], [9], [10], an iterative algorithm similar to N-mode SVD [8], [10], [22] is proposed for determining the minimum. More strictly speaking, as shown in Theorem 1, if $(U_1, \ldots, U_{k-1}, U_{k+1}, \ldots, U_n)$ are given, then the optimal U_k can be acquired by simple eigenvalue decomposition. So, after initializing $U_k|_{k=1}^n$ as identity matrices or arbitrary columnly-orthogonal matrices, the optimal $U_k, k = 1, \ldots, n$ can be acquired iteratively.

Theorem 1: If $(U_1, \ldots, U_{k-1}, U_{k+1}, \ldots, U_n)$ are given, the optimal U_k in (5) is composed of the leading m'_k eigenvectors of the covariance matrix $C^k = \sum_{i=1}^N X_i^k X_i^{k^T}$, where X_i^k is the mode-k flattened matrix of $\mathbf{X}_i^k$, i.e., $X_i^k \Longleftarrow_k \mathbf{X}_i^k$, and $\mathbf{X}_i^k$ is defined as

$$\mathbf{X}_i^k = \mathbf{X}_i \times_1 U_1^T \ldots \times_{k-1} U_{k-1}^T \times_{k+1} U_{k+1}^T \ldots \times_n U_n^T. \quad (6)$$

The proof of this theorem can be easily derived from [7], [9], and [10] as shown in the Appendix. The detailed procedure for acquiring the optimal solution is described in Procedure–1.

Fig. 2 illustrates one step of CSA with third-order tensors. Fig. 2(a) shows N original tensor objects of third order, where each cubic may represent a set of Gabor filtered images. Fig. 2(b) shows the new tensor objects after the mode-2 and mode-3 product operations with U_2^{t-1} and U_3^{t-1} respectively. Then each cubic is unfolded into a matrix with the mode-1 flattening operation, as shown in Fig. 2(c). The N matrices are concatenated into a new matrix, as in Fig. 2(d) because the covariance matrix of the concatenated matrix is equal to the covariance matrix $C^1 = \sum_{i=1}^N X_i^1 X_i^{1^T}$. Then, PCA is performed to acquire the eigenvectors U_1^t of C^1.

Given the sample set $\mathbf{X}_i \in \mathbb{R}^{m_1 \times m_2 \times \ldots \times m_n}$, $i = 1, \ldots, N$, the final lower dimensions $m'_k, k = 1, \ldots, n$, and the iteration number T_{max}:

1) Initialize U_k^0, $k = 1, \ldots, n$ as arbitrary columnly-orthogonal matrices[2] ;
2) For $t = 1, 2, \ldots, T_{max}$ Do
   a) For $k = 1, 2, \ldots, n$ Do
   - Compute $\mathbf{X}_i^k$: $\mathbf{X}_i^k = \mathbf{X}_i \times_1 U_1^{t^T} \ldots \times_{k-1} U_{k-1}^{t^T} \times_{k+1} U_{k+1}^{t-1^T} \ldots \times_n U_n^{t-1^T}$
   - Perform mode-k flattening of $\mathbf{X}_i^k$ to matrix X_i^k: $X_i^k \Longleftarrow_k \mathbf{X}_i^k$
   - Compute the covariance matrix C^k: $C^k = \sum_{i=1}^N X_i^k X_i^{k^T}$
   - Compute the first m'_k leading eigenvectors of C^k, $C^k U_k^t = U_k^t \Lambda_k$, namely the first m'_k eigenvectors constitute the column vectors of the matrix U_k^t,

   b) If $t > 2$ and $Tr[abs(U_k^{t^T} U_k^{t-1})]/m'_k > (1 - \epsilon)$ ($\epsilon = 0.001$ in our implementation), break[3];
3) Output the matrices $U_k = U_k^t$, $k = 1, \ldots, n$.

[2]If $U_k^T U_k = I$, then U_k is columnly-orthogonal matrix.
[3]$abs(\cdot)$ represents the absolute value for each element of a matrix.

Procedure–1. Procedure for CSA.

C. Algorithmic Analysis—Learnablity, Computational Complexity, and Objective Function Convergence

We discuss the properties of the proposed scheme in terms of learnability, computational complexity, and objective function convergence.

Learnability: Learnability is a problem in real applications such as face recognition that typically have small training sets and a large feature set. In such cases, lack of data leads to difficulties in finding the most representative subspace. In PCA, the size of the covariance matrix is $\prod_{k=1}^n m_k \times \prod_{k=1}^n m_k$ if a tensor is transformed into a vector. For most pattern recognition problems, a feature dimension of $\prod_{k=1}^n m_k$ is considered very large, since a substantial number of training samples would be needed

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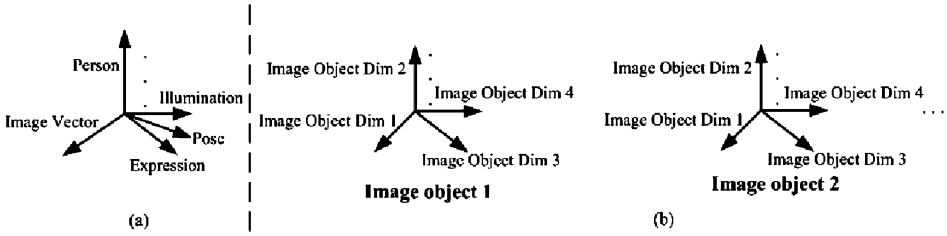


Fig. 3. Tensor composition in (a) Tensorface and (b) CSA.

to reliably acquire a representative subspace. In CSA, however, the step-wise covariance matrix C^k is of size $m_k \times m_k$, which is much smaller than that of PCA. The objects to be analyzed in CSA are the column vectors of the unfolded matrices, such that the length of an object is just m_k, which is much smaller than $\prod_{k=1}^{n} m_k$ in PCA. Moreover, the sample number in CSA is $N \prod_{i=1, i\neq k}^{n} m_k$, which is much larger than in PCA. With more training data and fewer feature dimensions, CSA can acquire a more representative subspace in comparison to PCA. For real face recognition applications, CSA is better able to discover a truly representative subspace, so it has the potential to map new test data into a more correct subspace and thus directly improve recognition performance. Furthermore, the number of available projection directions in CSA can be much larger than the sample number, which is not the case for most traditional algorithms. These properties directly elevate the performance of CSA above traditional algorithms in small sample size cases, and in fact CSA can work with even only one available sample. In addition, it also provides more useful information for subsequent supervised algorithms, such as LDA, to improve their performance.

Computational Complexity: For greater clarity, let us assume that the sample tensors are of uniform size for all dimensions, i.e., $m_i = m, i = 1, \ldots, n$. Also, we assume that the total number of samples N is comparable in magnitude to the feature dimension m^n. In this case, the complexity of PCA is $O(m^{3n})$.[2] If the images are of high resolution, e.g., more than 20 000 pixels, and the number of training images is of the same magnitude, it is impossible for current computers to conduct eigenvalue decomposition. However, in CSA, the complexity of calculating the covariance matrices is $O(nm^{n+1})$, and the complexity of general eigenvalue decomposition is $O(nm^3)$ for each loop, which is considerably lower than that of PCA. Although CSA has no closed-form solution and many loops are required for optimization, it is still much faster than PCA owing to its simplicity in each iteration loop.

Objective Function Convergence: Let $h(U_1^t, U_2^t, \ldots, U_n^t) = \sum_{i=1}^{N} \|\mathbf{X}_i \times_1 U_1^t U_1^{t^T} \ldots \times_n U_n^t U_n^{t^T} - \mathbf{X}_i\|^2$. From (5) and Procedure–1, we pursue a minimum at each substep such that the objective function is non-increasing

$$\begin{aligned} h\left(U_1^t, U_2^t, \ldots, U_n^t\right) &\geq h\left(U_1^{t+1}, U_2^t, \ldots, U_n^t\right) \\ &\geq h\left(U_1^{t+1}, U_2^{t+1}, \ldots, U_n^t\right) \geq \ldots \\ &\geq h\left(U_1^{t+1}, U_2^{t+1}, \ldots, U_n^{t+1}\right). \end{aligned} \tag{7}$$

[2]Although eigenvalue decomposition can be conducted on a covariance matrix of size $N \times N$ when N is much less than m^n, we can consider the complexity of PCA as $O(m^{3n})$ in the case that N is comparable in magnitude to m^n.

Since the objective function has a lower bound $h(U_1^t, U_2^t, \ldots, U_n^t) \geq 0$, it will converge to a local optimum.

III. Relation to Previous Works

In this section, we discuss the proposed CSA algorithm in relation to some previous techniques. We first discuss the difference between CSA and Tensorface [19], [21], and then examine its connection to PCA [18] and to several algorithms based on matrix representations [15], [26], [27], [30], [31].

A. CSA and Tensorface

Tensorface [19], [21] is a recently proposed dimensionality reduction algorithm for face recognition with pose, illumination and expression variations. It also utilizes higher order tensor decomposition [7], [9], [10] for data analysis similar to CSA. While Tensorface and CSA are both variants of N-mode SVD [8], [10], [22] from an algorithmic viewpoint, they are intrinsically different because of their basic tensor compositions. As shown in Fig. 3, a face image in Tensorface is treated as a vector, and the face image ensembles of all persons under different pose, illumination and expression are treated as a high-order tensor. However, in CSA an image object, such as a face, digital number, or other general image, is directly treated as a high-order tensor. In Fig. 3(b), different persons are represented by different tensors in CSA, for ease of unifying previous work such as [31] into our framework, as shown in Section III-C. Also, it is unnecessary to conduct dimensionality reduction along the person dimension for object recognition. We note that Person-Specific Tensorface [19], [20] also represents different persons as different tensors.

This difference in tensor composition differentiates Tensorface and CSA in several aspects. 1) Tensorface and CSA have completely different motivations. Tensorface specifically deals with gray-level image input, which is converted to a 1-D vector, and thus it can be classified as an *image-as-vector* representation. Tensorface aims to characterize the variations caused by *external* factors, such as pose, illumination and expression. We also note that Tensorface does not address Gabor-filtered images or video sequences as input; it is unclear how Tensorface can deal with such image data. In CSA, we directly represent one gray-level image as a second order tensor (a matrix), and one Gabor filtered image or one video sequence as a third order tensor, thus it can be classified as an *image-as-matrix* representation. CSA aims to use several subspaces to characterize the *internal* variations of an object, such as of rows, columns, and

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Gabor features. 2) Tensorface requires a large number of samples of different pose, illumination and expression for training, which is generally unavailable in real applications such as face recognition. *When the number of images per person is only one or the total number of images is a prime, Tensorface is strictly equivalent to PCA for gray-level image input*. In contrast, when each person has only a limited number of images or even just a single image, CSA is more effective than PCA, as analyzed in Section II-C. 3) Potentially, CSA could be extended to address both internal and external factors by constructing tensors with the first several dimensions representing width, height, Gabor features, etc., and the other dimensions encoding additional factors such as pose, illumination, and expression.

B. Connection To PCA

CSA can be viewed as a PCA with constraints that increase the learnability of a representative subspace. For the case of vector input data, these constraints fall away, such that CSA and PCA are equivalent.

Claim 1: CSA is a constrained version of PCA, i.e., the projection matrix is constrained as the Kronecker product of projection matrices.

Proof: Suppose the tensors $\mathbf{X}_i$ and $\mathbf{Y}_i$ are vectorized dimension-by-dimension as x_i^v and y_i^v. Since $\mathbf{Y}_i = \mathbf{X}_i \times_1 U_1^T \ldots \times_n U_n^T, i = 1, \ldots, N$, it is easy to show that $y_i^v = P^T x_i^v$ with

$$P = U_n \otimes U_{n-1} \otimes, \ldots, \otimes U_1 \in \mathbb{R}^{\prod_{k=1}^{n} m_k \times \prod_{k=1}^{n} m_k'}$$

where $\otimes$ is the Kronecker product with $A \otimes B = [A_{ij}B]$. The objective function of CSA can also be rewritten as that of PCA:

$$\sum_{i=1}^{N} \|\mathbf{X}_i \times_1 U_1 U_1^T \ldots \times_n U_n U_n^T - \mathbf{X}_i\|^2 = \sum_{i=1}^{N} \|PP^T x_i^v - x_i^v\|^2. \quad (8)$$

Similar to conventional PCA, the projection matrix P is columnly orthogonal, yet P is constrained to be in the form of Kronecker products of multiple matrices. Hence, it can be considered as a constrained version of PCA. □

Claim 2: For vector input ($n = 1$), CSA and PCA are equivalent.

Proof: When $n = 1$, a tensor can be directly represented as a vector $x_i = \mathbf{X}_i$. The objective function of CSA is then

$$\sum_{i=1}^{N} \|\mathbf{X}_i \times_1 U_1 U_1^T - \mathbf{X}_i\|^2 = \sum_{i=1}^{N} \|U_1 U_1^T x_i - x_i\|^2 = -NTr(U_1^T C U_1) + c_1 \quad (9)$$

where c_1 is a constant and $C = \frac{1}{N}\sum_{i=1}^{N} x_i x_i^T$ is the covariance matrix of the conventional PCA algorithm if we assume that the samples are centered before applying PCA. Therefore, for the case of first-order tensor data, CSA and the traditional PCA algorithm are equivalent. □

C. Connection To 2-D Matrix Based Algorithms [15], [26], [27], [30], [31]

We first describe the connection between CSA and 2-D matrix based algorithms, and then show that the work in [26], [27], [30], and [31] can be considered as a special case of CSA.

Similar to Claim 1, the work in [15], [27], and [31] can be considered as a constrained PCA. In [15], each column of the projection matrix $P = [p_1, p_2, \ldots, p_d]$ has the form of a Kronecker product of unitary vectors: $p_j = v_j \otimes u_j \; u_j \in \mathbb{R}^m, v_j \in \mathbb{R}^n, \|u_j\| = 1, \|v_j\| = 1, \; \forall \; j$. In [27] and [31], the projection matrix is $P = U \otimes V, U \in \mathbb{R}^{m \times m'}, V \in \mathbb{R}^{n \times n'}$. So the difference among [15], [27], and [31] is that they use different constraints. Furthermore, the objective functions in [15], [27], and [31] do not have closed-form solutions, so they proposed similar iterative algorithms to determine a minimum.

As shown in [27] and [31], 2DPCA [30] is a special case of [27] and [31] with U as the identity matrix. Moreover, as shown in [26], 2DPCA can be considered as a special PCA with the row vector as the analyzed object, i.e., 2DPCA only removes the redundancy among the row vectors and ignores the redundancy among the column vectors, thus 2DPCA requires more coefficients for image representation than PCA [30]. To remove the redundancy among both row and column vectors, PIMC [26] was proposed to conduct PCA with row and column vectors of the image as the object in each step. From an algorithmic viewpoint, it can be considered as a special case of [27] and [31] having only the first iteration.

Claim 3: Coupled Subspace Analysis [27] or Low Rank Approximation of Matrices [31] is a special case of the CSA algorithm with $n = 2$.

Proof: Similar to the Proof of Claim 2, we assume that the tensor $\mathbf{X}_i$ can be directly represented as a matrix X_i. Thus, the objective function of CSA can be rewritten as

$$\sum_{i=1}^{N} \|\mathbf{X}_i \times_1 U_1 U_1^T \times_2 U_2 U_2^T - \mathbf{X}_i\|^2 = \sum_{i=1}^{N} \|U_1 U_1^T X_i U_2 U_2^T - X_i\|^2. \quad (10)$$

This objective function is the same as that in (3), and thus CSA [27] or LRAM [31] is a special case of CSA with $n = 2$. □

Furthermore, 2DPCA [30] is also a special case of CSA with $U_2 = I$ and $n = 2$, and PIMC [26] is also a special case of CSA with $n = 2$ and maximum iteration number $T_{\max} = 1$.

IV. Experiments

We first compare CSA with conventional PCA on object reconstruction, and then we compare them on two object recognition tasks: face recognition and digital number recognition.

A. Database Description

We use the following four databases in our experiments.

FERET face database [14]: The FERET Fa/Fb database consists of 1195 persons, with each person having two images. The gray-level images are all normalized to size 41×27, aligned by fixing the locations of the two eyes, and preprocessed with histogram equilibrium.

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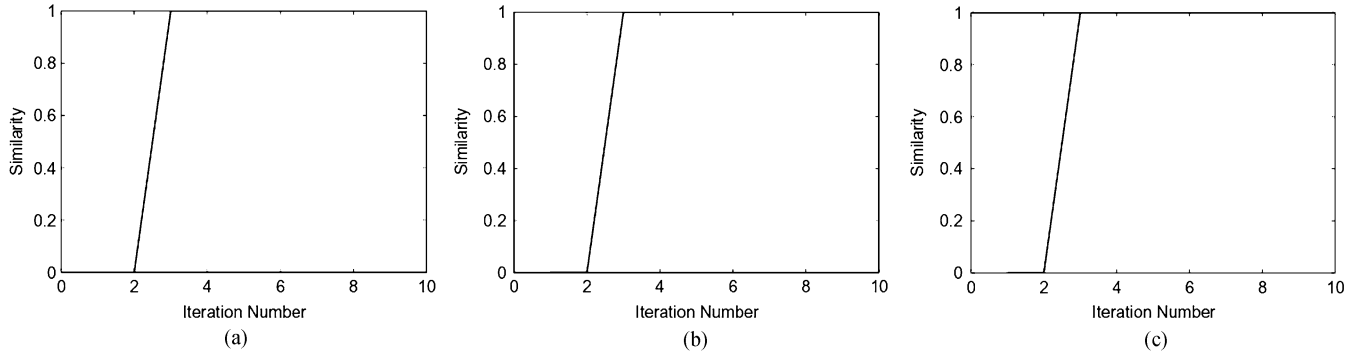


Fig. 4. Convergence characteristics of CSA on the ORL database with $m'_1 = m'_2 = m'_3 = 10$. The vertical axis indicates the similarity of two successive projection matrices, and the horizontal axis gives the number of iterations.

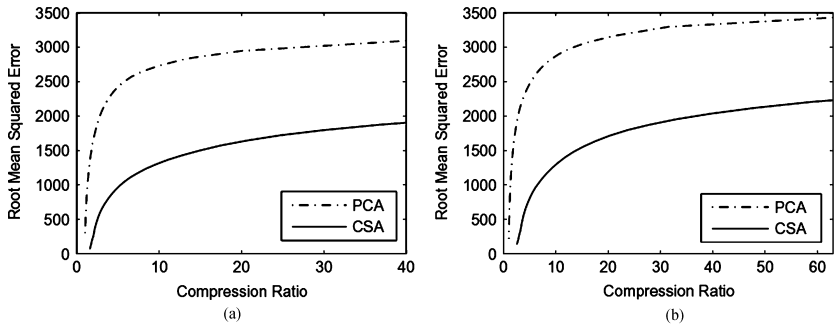


Fig. 5. Comparison of CSA and PCA for object reconstruction on the ORL and CMU PIE-1 databases.

ORL Face Database [13]: The ORL database [13] contains 400 images of 40 individuals. All images are grayscale and normalized to a resolution of 56×46 pixels. A preprocessing step of histogram equilibrium is applied.

CMU PIE Face Database [16]: In this experiment, two subdatabases are chosen from the CMU PIE database. In the first subdatabase, referred to as PIE-1, five near frontal poses (C27, C05, C29, C09, and C07) and illumination indexed as 08 and 11 are used. Another subdatabase PIE-2 consists of the same five poses as in PIE-1, and another two illuminations 10 and 13 are used. The PIE-2 database is more challenging than PIE-1 for classification because of the flash position. Due to incompleteness of data, only 63 persons are used in this work with each person having ten images. All the gray-level images are normalized to a resolution of 64×64 pixels with fixed locations of the two eyes. Histogram equilibrium is used for preprocessing.

MNIST Digital Number Database [11]: The first half of the test set of the MNIST database is used in this experiment, which consists of images of 5 000 handwritten numbers ("0"–"9") with each digital number having 500 images. The image size is 28×28.

B. Object Reconstruction

In this experiment, we consider 3-D Gabor feature input. We first use the ORL database to illustrate the algorithmic convergence, and then we compare CSA and PCA for object reconstruction with the ORL and CMU PIE-1 databases. We utilize only *one* image per person in order to highlight the difference between CSA and Tensorface, which needs *several* images to construct the tensor and in this case is strictly equivalent to PCA. We extracted 40 Gabor features with five different scales and eight different orientations [25], so each image is then represented as a third order tensor of size $56 \times 46 \times 40$ for the ORL database and $64 \times 64 \times 40$ for the CMU PIE-1 database.

Fig. 4 illustrates the convergence characteristics of CSA, in which the horizontal axis indicates the number of iterations, and the vertical axis is the similarity of three successively estimated projection matrices, i.e., $Tr[\mathrm{abs}(U_k^{tT} U_k^{t-1})]/m'_k, k = 1, 2, 3$, and $m'_1 = m'_2 = m'_3 = 10$. From Fig. 4, it is clear that CSA often converges after about three iterations. We have observed the same trend in other databases as well as with different values of m'_k.

In [28], we reported that the reconstructed images of CSA are better than those of PCA in both visual quality and similarity to a video sequence constructed from the CMU PIE database by ordering the 13 poses along the time dimension. Here, we support those qualitative judgements with a quantitative evaluation. Fig. 5 illustrates the *root mean squared error* (RMSE) with different *compression ratios* (CRs) for CSA and PCA. We use the same definition of CR as in [31]: $Nm_1m_2m_3/s$ with s as the number of scales required to represent the data. Note that as shown in [23] and [31], we also need to store the projection matrices. So for PCA, we have $s = d(N + m_1m_2m_3)$ with d as the dimension after reduction, and for CSA we have $s = Nr^3 + r(m_1 + m_2 + m_3)$, taking $m'_1 = m'_2 = m'_3 = r$ for simplicity as suggested in [23] and [31]. The RMSE is defined as $\sqrt{\sum_{i=1}^{N} ||PP^T x_i - x_i||^2/N}$ for PCA and $\sqrt{\sum_{i=1}^{N} ||\mathbf{X}_i^R - \mathbf{X}_i||^2/N}$ for CSA. While PCA achieves the optimal reconstruction when the image object

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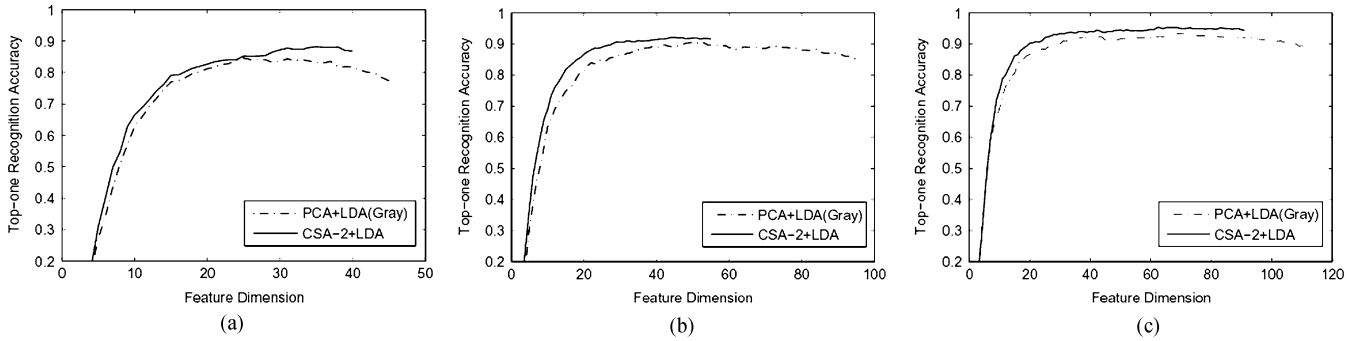


Fig. 6. Comparison of $\mathrm{PCA}+\mathrm{LDA}$ (gray) and $\mathrm{CSA\text{-}2}+\mathrm{LDA}$ on the FERET Fa/Fb database. For $\mathrm{PCA}+\mathrm{LDA}$ (gray), the lower dimensionality of the PCA step are set to 45, 95, and 112, respectively. For $\mathrm{CSA\text{-}2}+\mathrm{LDA}$, the lower dimensionality in CSA-2 step are set to 10×4, 11×5, and 13×7, respectively.

TABLE I
TOP-ONE RECOGNITION RATES (%) OF $\mathrm{CSA\text{-}2}+\mathrm{LDA}$ AND $\mathrm{PCA}+\mathrm{LDA}$ (GRAY) ON THE FERET FA/FB DATABASE

Algorithm	T / 50	T / 100	T / 200
PCA+LDA (Gray)	84.6 (45,25)	90.6 (95,40)	93.4 (112,69)
CSA-2+LDA	88.2 ($10\times 4, 35$)	92.2 ($11\times 5, 44$)	95.2 ($13\times 7, 62$)

TABLE II
TOP-ONE RECOGNITION ACCURACY (%)OF DIFFERENT ALGORITHMS ON THE ORL DATABASE

Algorithm	ORL
PCA+LDA (Gray)	82.2 (30,18)
CSA-2+LDA	84.1 ($11\times 2, 16$)
PCA (Gabor)	86.6 (76)
CSA-3	94.4 ($15\times 1\times 26$)
PCA+LDA (Gabor)	87.2 (33,25)
CSA-3+LDA	90.9 ($4\times 2\times 3, 23$)

is converted to a 1-D vector, from Fig. 5 it is clear that CSA outperforms PCA in object reconstruction. Similar trends are observed on other databases, which demonstrates that CSA can better remove the redundancies among different feature dimensions.

C. Object Recognition

In this experiment, we consider both 2-D grayscale features and 3-D Gabor feature input, referred to as CSA-2 and CSA-3. For 2-D grayscale feature input, our previous work [27] as well as the work in [31] have demonstrated that CSA-2 outperforms 2DPCA and PCA in face recognition. Here, we compare $\mathrm{CSA\text{-}2}+\mathrm{LDA}$ with $\mathrm{PCA}+\mathrm{LDA}$ [1] to examine their performance as a first step for dimensionality reduction. For all the experiments, the nearest neighbor classifier is applied after dimensionality reduction to obtain a final decision.

For the LDA related algorithms $\mathrm{PCA}+\mathrm{LDA}$, $\mathrm{CSA\text{-}2}+\mathrm{LDA}$ and $\mathrm{CSA\text{-}3}+\mathrm{LDA}$, to obtain a fair comparison and avoid singularity problems in LDA [1], [4], we set the reduced dimension after PCA, CSA-2, and CSA-3 to be no larger than $N-N_c$, where N_c is the total number of classes. For the CSA-2 and CSA-3 related algorithms, we observe that $T_{\max}=3$ is sufficient for algorithmic convergence, and we use this value in this work. For all the algorithms, we report the best results, with the optimal parameter configurations given in brackets. More specifically, for PCA and CSA-3 the numbers in brackets indicate the optimal lower dimension; for $\mathrm{PCA}+\mathrm{LDA}$, $\mathrm{CSA\text{-}2}+\mathrm{LDA}$ and $\mathrm{CSA\text{-}3}+\mathrm{LDA}$, the first number in brackets is the optimal lower dimension in the PCA, CSA-2, or CSA-3 step, and the second one is the optimal lower dimension in the LDA step.

FERET Database (CSA-2): To compare performances for different training set sizes, we consider different numbers of persons in the training set, denoted as T/q in Table I. Specifically, we select the first q persons for training, and the remaining persons are used for testing. We also use the images from the Fa set as gallery images, and Fb is used as the probe set. Fig. 6 plots recognition rates with respect to the LDA dimension. As shown in Table I and Fig. 6, $\mathrm{CSA\text{-}2}+\mathrm{LDA}$ outperforms $\mathrm{PCA}+\mathrm{LDA}$ (Gray), especially for smaller sample sizes.

ORL Database (CSA-2 and CSA-3): In this experiment, two images per person are randomly selected for the training and the gallery sets, and the remaining eight images compose the probe set. For CSA-3, we extracted 40 Gabor features with five different scales and eight different directions at down-sampled positions [25]; so each image is then represented as a third-order tensor of size $28\times 23\times 40$. The experimental results are shown in Table II, and Fig. 7 illustrates the face recognition rates when using different numbers of features. From it, we can clearly observe that: 1) CSA-3 performs better than PCA (Gabor); 2) $\mathrm{CSA\text{-}2}+\mathrm{LDA}$ and $\mathrm{CSA\text{-}3}+\mathrm{LDA}$ outperform $\mathrm{PCA}+\mathrm{LDA}$ (Gray) and $\mathrm{PCA}+\mathrm{LDA}$ (Gabor), respectively. We notice that the results of $\mathrm{CSA\text{-}3}+\mathrm{LDA}$ are worse than CSA-3, and the improvement of $\mathrm{PCA}+\mathrm{LDA}$ (Gabor) over PCA (Gabor) is very limited. This observation can be explained by the small number of available samples, which are insufficient to characterize the data distribution; LDA fails because of overfitting in this case.

CMU PIE Database (CSA-2 and CSA-3): In this experiment, three images per person are randomly selected as the gallery set, and the remaining seven images per person are used as the probe set. For the third-order tensor representation, we extract 40 Gabor features as in [25] with five different scales and eight different directions at down-sampled positions, and then encode each image as a third-order tensor of size $16\times 16\times 40$. The experimental results are shown in Table III, and Fig. 8 illustrates the face recognition rates when using different numbers of features for the PIE-1 database. From it, we have observations similar to that of the ORL database, that the CSA-2 and CSA-3 based algorithms outperform the related versions of PCA.

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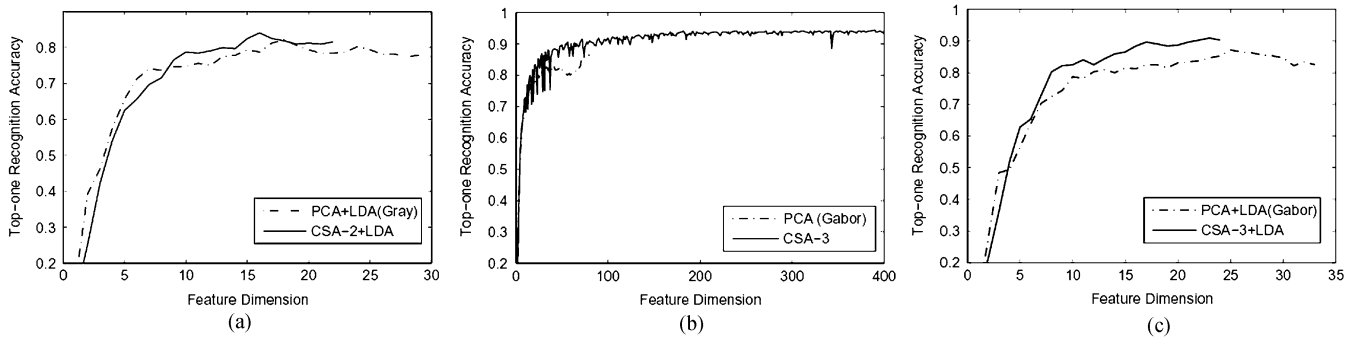


Fig. 7. Comparison of different algorithms on the ORL database. (a) For $\mathrm{PCA} + \mathrm{LDA}$ (gray) and $\mathrm{CSA\text{-}2} + \mathrm{LDA}$, the lower dimensionality in the PCA and CSA-2 step are set to 30 and 11×2, respectively. (b) For CSA-3, the feature number is the product of the reduced dimensions. (c) For $\mathrm{PCA} + \mathrm{LDA}$ (Gabor) and $\mathrm{CSA\text{-}3} + \mathrm{LDA}$, the lower dimensionality in the PCA and CSA-3 step are set to 33 and $4 \times 2 \times 3$, respectively.

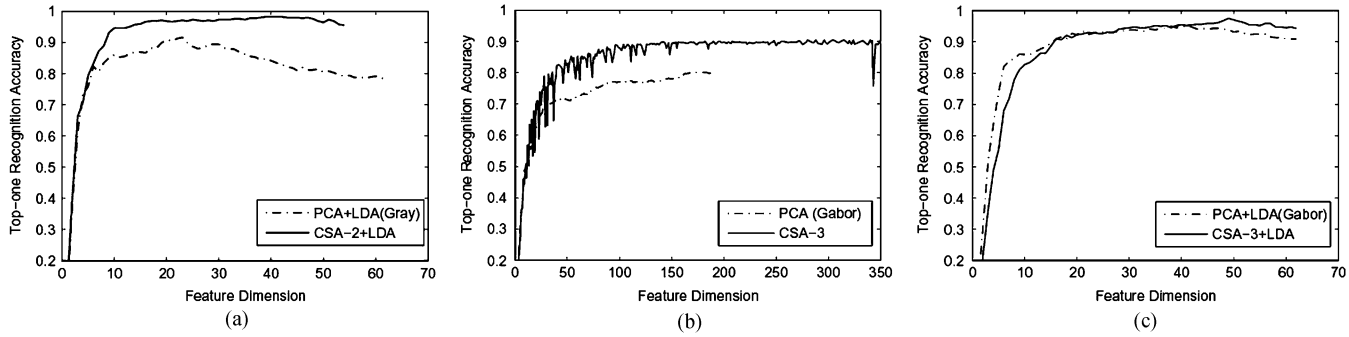


Fig. 8. Comparison of different algorithms on the CMU PIE-1 database. (a) For $\mathrm{PCA} + \mathrm{LDA}$ (gray) and $\mathrm{CSA\text{-}2} + \mathrm{LDA}$, the lower dimensionality in the PCA and CSA-2 step are set to 67 and 9×6, respectively. (b) For CSA-3, the feature number is the product of the reduced dimensions. (c) For $\mathrm{PCA} + \mathrm{LDA}$ (Gabor) and $\mathrm{CSA\text{-}3} + \mathrm{LDA}$, the lower dimensionality in the PCA and CSA-3 step are set to 74 and $3 \times 2 \times 12$, respectively.

TABLE III
TOP-ONE RECOGNITION ACCURACY (%) OF DIFFERENT ALGORITHMS ON THE CMU PIE DATABASE

Algorithm	CMU PIE-1	CMU PIE-2
PCA+LDA (Gray)	91.6 (67,23)	73.0 (65,16)
CSA-2+LDA	98.2 ($9 \times 6, 39$)	81.9 ($10 \times 5, 21$)
PCA (Gabor)	80.1 (168)	42.0 (189)
CSA-3	90.5 ($10 \times 1 \times 34$)	59.4 ($14 \times 1 \times 33$)
PCA+LDA (Gabor)	95.2 (74,41)	75.5 (65,62)
CSA-3+LDA	97.5 ($3 \times 2 \times 12, 49$)	84.8 ($6 \times 1 \times 13, 45$)

TABLE IV
TOP-ONE RECOGNITION ACCURACY (%) OF PCA (GABOR) AND CSA-3 ON THE MNIST DATABASE

Algorithm	$G30$ / $P470$	$G40$ / $P460$
PCA (Gabor)	88.5 (60)	90.5 (50)
CSA-3	92.8 ($5 \times 9 \times 18$)	92.4 ($15 \times 5 \times 13$)

MNIST Database (CSA-3): To further highlight the broader application of CSA in comparison to Tensorface, which is intended for collections of face ensembles, we also apply our algorithm to digital number recognition. These experiments are labeled as Gm/Pn, which indicates that m images per digital number are randomly selected for training and the remaining n images are used for testing. Since Gabor features have shown better performance than gray-level features [5] in this task, we extracted 40 Gabor features with five different scales and eight different directions as in [25], and then encoded each image as a third order tensor of size $28 \times 28 \times 40$.

We notice that digital number recognition is a typical "large sample size" problem, but it still suffers from high dimensionality, especially when Gabor features are considered. We do not compare our algorithm with other digital number recognition schemes, since the purpose of this experiment is to compare different object representations in recognition accuracy. The experiments are reported in Table IV. It can be observed that CSA-3 outperforms PCA (Gabor) in classification accuracy, and the result again demonstrates that tensor based representations can benefit classification algorithms.

Discussions: From the above experiments, we have the following observations. 1) For unsupervised experiments, CSA-3 is consistently better than PCA (with Gabor features), which demonstrates that algorithms based on tensor representations can map new test data into a more correct subspace when only limited training samples are available, which directly improves recognition performance in small sample size cases. 2) $\mathrm{CSA\text{-}2} + \mathrm{LDA}$ and $\mathrm{CSA\text{-}3} + \mathrm{LDA}$ outperform $\mathrm{PCA} + \mathrm{LDA}$ (Gray) and $\mathrm{PCA} + \mathrm{LDA}$ (Gabor), respectively, which demonstrates that CSA-2 and CSA-3 can provide more useful information to improve the performance of the subsequent supervised algorithm LDA. 3) From the results on the FERET and MNIST databases, the improvement of CSA over PCA is more pronounced when the number of training samples is small, which is consistent with the conclusion in [15]. 4) The results of $\mathrm{PCA} + \mathrm{LDA}$ (Gabor) are better than that of $\mathrm{PCA} + \mathrm{LDA}$ (Gray), and $\mathrm{CSA\text{-}3} + \mathrm{LDA}$ outperforms $\mathrm{CSA\text{-}2} + \mathrm{LDA}$ in most cases, which demonstrates that Gabor

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features can improve recognition performance in comparison to grayscale features. 5) In our experiments, we observed that for different databases the best results may be from different feature extraction methods. Also, it is possible that some algorithms are better with lower dimensions compared with other algorithms, yet not so good in the overall result. A possible explanation is that the final results may be affected by many factors, including feature set (grayscale values or Gabor filtered images) and distribution consistency of the training set and testing set. How to select the best feature extraction method and the optimal reduced dimensions are still open questions.

V. Conclusion

In this paper, we have proposed a new algorithm, CSA, for optimal object reconstruction with image objects represented as higher order tensors. First, we proposed the optimal reconstruction criterion for higher order tensor objects, encoded in multiple concurrent subspaces. An iterative procedure is presented to sequentially learn these concurrent subspaces. By representing image objects in their intrinsic form and order rather than reorganizing the high-dimensional data into 1-D vectors, CSA outperforms the classical PCA in terms of object reconstruction and object recognition. To the best of our knowledge, this is the first work to address the optimal object reconstruction problem with the objects represented as high-order tensors of arbitrary order. Our current solution is a variant of N-mode SVD [8], [10], [22]. Recent work [12] has shown that Parallel Factors Decomposition or Canonical Decomposition, which decompose a tensor as a linear combination of a minimal number of rank-1 tensors, achieve better performances than N-mode SVD in content-based indexing and retrieval, so in future work we plan to apply Parallel Factors Decomposition to further improve performance. Another interesting research direction is to develop a new algorithm for lower-rank tensor approximation with a globally optimal solution.

Appendix

Lemma-1: For arbitrary matrices A and B, we have $\mathbf{X} \times_{k_1} A \times_{k_2} B = \mathbf{X} \times_{k_2} B \times_{k_1} A (k_1 \neq k_2)$, and $(\mathbf{X} \times_k A) \times_k B = \mathbf{X} \times_k (BA)$.

Lemma-2: $\|\mathbf{X} \times_k U_k^T\|^2 = \|U_k^T X^k\|^2$, where X^k is the mode-k flattened matrix of $\mathbf{X}$, i.e., $X^k \Longleftarrow_k \mathbf{X}$.

Lemma-3: If matrix U_k satisfies $U_k^T U_k = I$, we have $\|\mathbf{X} \times_k U_k U_k^T\|^2 = \|\mathbf{X} \times_k U_k^T\|^2$.

Lemma-4: For an arbitrary matrix X, we have $\|X\|^2 = Tr(X^T X) = Tr(XX^T)$. **A.**

Proof of Theorem-1.

Proof: We denote $U_j|_{j=1,j\neq k}^n$ as $(U_1, \ldots, U_{k-1}, U_{k+1}, \ldots, U_n)$, and denote the objective function as $f(U_k)$ when the other projection matrices $U_j|_{j=1,j\neq k}^n$ are given. We note that $U_j|_{j=1}^n$ are columnly-orthogonal matrices, i.e., $U_k^T U_k = I, k = 1, \ldots, n$.

First, according to *Lemma-3* and *Lemma-1*, it is easy to prove that

$$
\begin{aligned}
f(U_k) &= \sum_{i=1}^{N} \|\mathbf{X}_i \times_1 U_1 U_1^T \ldots \times_n U_n U_n^T - \mathbf{X}_i\|^2 \\
&= \sum_{i=1}^{N} \|\mathbf{X}_i \times_1 U_1 U_1^T \ldots \times_n U_n U_n^T - \mathbf{X}_i \times_n U_n U_n^T\|^2 \\
&\quad + \|\mathbf{X}_i \times_n U_n U_n^T - \mathbf{X}_i\|^2 \\
&= \cdots \\
&= \sum_{i=1}^{N} \|\mathbf{X}_i \times_j U_j^T|_{j=1,j\neq k}^n \times_k \\
&\quad \times U_k U_k^T - \mathbf{X}_i \times_j U_j^T|_{j=1,j\neq k}^n\|^2 \\
&\quad + \|\mathbf{X}_i \times_j U_j U_j^T|_{j=1,j\neq k}^n - \mathbf{X}_i\|^2.
\end{aligned}
$$

The first equality comes from the fact that the matrix is columnly orthogonal. The latter part of the objective function is constant, denoted here as $c_2 = \sum_{i=1}^{N} \|\mathbf{X}_i \times_j U_j U_j^T|_{j=1,j\neq k}^n - \mathbf{X}_i\|^2$.

According to the definition in (6) and *Lemma-2*, we can rewrite $f(U_k)$ as

$$
\begin{aligned}
f(U_k) &= \sum_{i=1}^{N} \|\mathbf{X}_i^k \times_k U_k U_k^T - \mathbf{X}_i^k\|^2 + c_2 \\
&= \sum_{i=1}^{N} \|U_k U_k^T X_i^k - X_i^k\|^2 + c_2.
\end{aligned}
$$

Then, we can obtain the optimal projection matrix U_k with the given $U_j|_{j=1,j\neq k}^n$ as follows:

$$
\begin{aligned}
U_k &= \arg\min_{U_k} \sum_{i=1}^{N} \|U_k U_k^T X_i^k - X_i^k\|^2 \\
&= \arg\min_{U_k} \sum_{i=1}^{N} Tr\left[\left(U_k U_k^T X_i^k - X_i^k\right)^T \right. \\
&\quad \left. \times \left(U_k U_k^T X_i^k - X_i^k\right),\right] \\
&= \arg\min_{U_k} \left[\sum_{i=1}^{N} Tr\left(-U_k^T X_i^k X_i^{k^T} U_k\right) + c_3\right] \\
&= \arg\max_{U_k} Tr\left(U_k^T \sum_{i=1}^{N} X_i^k X_i^{k^T} U_k\right).
\end{aligned}
$$

The second step follows from *Lemma-4*. In the third step, the constant is $c_3 = \sum_{i=1}^{N} Tr(X_i^k X_i^{k^T})$. Thus, we prove that the optimal U_k with the given $U_j|_{j=1,j\neq k}^n$ is the leading m_k' eigenvectors of the covariance matrix $\sum_{i=1}^{N} X_i^k X_i^{k^T}$.

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